

AnalystLab Africa
Data Analytics Internship — Week 8 Capstone Project

Closing the Power Gap

Electrification Access Across Sub-Saharan Africa, 1990–2024

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Data source: World Bank World Development Indicators (WDI)
Tool: Microsoft Power BI

1. Objective / Problem Statement

Electricity access is one of the clearest markers of infrastructure development, yet national averages routinely hide a sharper divide: the gap between what a country's cities enjoy and what its rural population goes without. This project asks a specific, answerable question:

How has electricity access evolved across Sub-Saharan Africa since 1990, and which countries still show the widest gap between rural and urban access?

The goal is not simply to report that a gap exists, but to quantify it, rank it by country, track its trajectory over time, and identify where it is closing fastest — and where it is not closing at all.

2. Dataset Description

This analysis uses the World Bank's World Development Indicators (WDI), the institution's flagship collection of global development statistics, covering over 200 countries and territories across economy, health, education, environment, and infrastructure themes.

Source and Scope

- **Source file:** WDI bulk CSV download (WDICSV.csv), supplemented by WDIcountry.csv for regional classification.
- **Original size:** 396,970 rows across 1,498 indicators and 265 countries and regional aggregates, spanning 1960–2024.
- **Indicators selected:** EG.ELC.ACCS.ZS (Access to electricity, % of population), and its rural (EG.ELC.ACCS.RU.ZS) and urban (EG.ELC.ACCS.UR.ZS) breakdowns.
- **Geographic scope:** 48 countries classified by the World Bank as Sub-Saharan Africa.
- **Time scope:** 1990–2024. Electricity access is not reliably reported for this indicator before 1990; earlier years were excluded rather than imputed.

3. Data Cleaning Process

Every cleaning step below was applied deliberately, documented as it was performed, and — where possible — checked against the original source data rather than assumed correct.

1. Filtered the source file from 396,970 rows down to 795, keeping only the three electricity-access indicators (by an “Indicator Code begins with EG.ELC.ACCS” filter).
2. Reshaped the data from wide format (one column per year, 1960–2024) into long format (one row per country-indicator-year). This step automatically excluded cells with no reported value — primarily years before 1990 — rather than requiring a separate null-removal step.
3. Joined the country reference table on country code to attach each row's World Bank region. This simultaneously removed 47 non-country aggregate rows (e.g. “World,” “High income”), which carry no region classification of their own.
4. Filtered to Region = Sub-Saharan Africa, isolating the 48 countries in scope.

5. Checked for duplicate country-indicator-year records at three separate points in the pipeline (source rows, post-reshape, and post-pivot). None were found at any stage — consistent with WDI being a curated institutional dataset rather than raw transactional data, and confirmed rather than assumed.
6. Pivoted the three indicator rows into three columns (Total, Rural, and Urban Access), producing one row per country-year. This simplifies the urban-rural comparison used throughout the analysis.
7. Identified and corrected a data anomaly: Chad's reported rural electricity access jumped from 0.4% in 2023 to 100% in 2024 — inconsistent with a decade of readings under 5%, and with no explanatory footnote in WDI's own metadata. The single cell was set to blank rather than replaced with an estimated value, to avoid fabricating a figure that isn't known.

The final cleaned dataset contains 1,400 rows across 7 columns (Country, Country Code, Year, Region, Total/Rural/Urban Access), covering 48 countries and 1990–2024. Rural access retains genuine reporting gaps for roughly 199 of the 1,400 country-year rows, and Total access for 15; Urban access has complete coverage. These gaps were left blank rather than estimated, and are treated as a known limitation rather than resolved artificially (see Section 8).

4. Analysis Methodology

The cleaned dataset was loaded into Power BI, where nine DAX measures were built to move beyond simple totals and answer the driving question directly:

- **Access Gap / Latest Access Gap:** the urban-minus-rural access percentage, calculated for each country's most recently reported year rather than blended across decades of history.
- **Rolling 3-Year Average:** a trailing average built with an explicit year-window filter, since the dataset's annual grain does not warrant Power BI's built-in date-hierarchy time intelligence.
- **Gap Rank:** a RANKX measure ordering all 48 countries by current gap size, largest first.
- **Year-over-Year Change:** the most recent year's access level minus the prior year's, isolating genuine annual movement.
- **Supporting measures:** current regional access level, country count, and latest rural/urban access levels used in the country-comparison chart.

Each of these measures was individually verified against the raw source figures during development — for example, Angola's rolling average was hand-checked against its 2022–2024 values (48.5%, 51.1%, 55.5%) before being trusted in the dashboard.

Dashboard Design

The dashboard is built as a single, coherent page rather than a collection of unrelated charts, anchored on the driving question above:

- Five KPI cards giving the regional headline numbers at a glance.
- A choropleth map of Sub-Saharan Africa, shaded by each country's current access gap.
- A ranked bar chart of the ten countries with the widest current gap.
- A rolling-average trend line, filterable by country, showing trajectories over time.
- A rural-vs-urban comparison chart for the same ten countries, showing the two raw figures whose difference produces the headline gap.

- Country and Year slicers, plus two bookmark-linked buttons offering one-click preset comparisons (“Widest Gap Countries” vs. “Major Economies”).

5. Key Findings

Regional Headline Numbers (Latest Reporting Year)

Metric (latest reporting year)	Value
Regional avg. total electricity access (latest year)	58.23%
Regional avg. urban–rural access gap (latest year)	40.39 pts
Year-over-year change (regional)	+2.10 pts
3-year rolling average access level	56.09%
Countries covered	48

Widest Current Urban–Rural Gaps

Rank	Country	Urban–Rural Gap (pts)
1	Mauritania	94.20
2	Equatorial Guinea	84.20
3	Angola	76.70
4	Burkina Faso	75.10
5	Mozambique	67.50

Beyond the headline numbers, three specific findings stand out:

- **Progress is real but uneven.** Angola's total access rose from 45.6% in 2019 to 55.5% in 2024 — a genuine, steady year-on-year climb — while simultaneously carrying the third-widest urban-rural gap in the region (76.70 points). A country can be improving and deeply unequal at the same time.
- **The gap is driven by near-total rural exclusion in specific countries, not gradual lag.** In Mauritania, urban access sits near 94% while rural access is close to zero — not a modest shortfall, but a near-complete absence of rural electrification alongside a largely electrified capital.
- **Even large, well-known economies show a substantial divide.** Nigeria, one of the region's largest economies, reports 62.5% overall access — built from 85.5% urban access against just 23.5% rural access.

6. Insights

Three insights follow directly from the findings above:

- **A rising regional average can mask a widening domestic divide.** The region's overall access level is climbing (+2.10 points year-over-year), but this headline trend is compatible with individual countries — like Mauritania and Equatorial Guinea — sitting far above the average gap. National progress and rural equity are two different things worth tracking separately, not one number standing in for both.
- **The countries most in need of attention are not always the poorest overall.** Several of the widest-gap countries are not the region's lowest-access countries in total — they are countries where urban electrification has outpaced rural electrification specifically, which points to a targeting problem (where investment goes) rather than only a resourcing problem (how much investment exists).
- **Data completeness itself tracks the same divide it measures.** Rural access figures are missing far more often than urban or total figures across the dataset. The same infrastructure and monitoring gaps that leave rural areas under-electrified likely also leave them under-measured — meaning the true rural gap may be even harder to see clearly than these figures already suggest.

7. Recommendations

- **Prioritize rural-specific investment in the five widest-gap countries** — Mauritania, Equatorial Guinea, Angola, Burkina Faso, and Mozambique — where national averages are most likely to overstate real rural progress.
- **Separate “improving but unequal” countries from “stagnant and unequal” countries in resource allocation.** Angola's steady year-over-year gains suggest continuity of current investment may be the right call; a country with a wide gap and flat or negative year-over-year change likely needs a change in approach, not more of the same.
- **Invest in rural data collection alongside rural infrastructure.** Given the coverage gaps identified in Section 3, improving how rural electrification is measured is a comparatively low-cost, high-leverage step — decision-makers cannot manage what isn't reliably reported.
- **Adopt basic anomaly checks before publishing or acting on updated figures.** The Chad data correction in Section 3 was caught only through manual cross-checking; a simple year-over-year jump threshold would flag this kind of implausible figure automatically.

8. Conclusion

Sub-Saharan Africa has made genuine, measurable progress in electricity access since 1990 — but that progress is unevenly distributed, and a country's overall access level can conceal a rural population still largely without power. The urban-rural gap, not just the national average, is the more direct metric for understanding where electrification is actually reaching people and where it is not. Closing that specific gap — rather than only raising headline access figures — is the more meaningful lever for genuine energy equity across the region.

With more time or data, the next steps worth pursuing would be: population-weighting the regional averages (this analysis treats all 48 countries equally regardless of population size); correlating the access

gap against other WDI indicators such as poverty and education to quantify downstream impact; and projecting current year-over-year rates forward to estimate when — if ever — specific high-gap countries would close the divide under present trends.